

SENSI

a sensorial - comfort copilot for architectural layouts

the number isn't the lesson → the edges are

△ thermal ○ visual ~ acoustic □ spatial ≈ olfactory : tactile

A space is a conversation between your senses

Soften a wall and the room goes quiet. Open a window and it smells fresh — but gets louder. Every change echoes from one sense to the next.

Sensi reads that conversation, and teaches it.

“The overall score is the least interesting part. The lesson lives in the edges — how a change to one sense ripples to another.”

what it does · reads a plan → scores 6 senses against your persona → finds conflicts → suggests & edits → shows the ripple

No two people feel a room the same

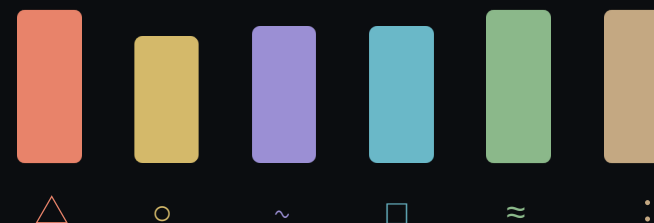
A calm introvert and a buzzy extrovert want opposite things from the same living room. So Sensi never scores against an average — your onboarding becomes a set of weights, and those weights decide what “comfortable” means for you. Care more about a sense, and Sensi flags it sooner.

$$t = \text{clamp}(0.35 + 0.40 \cdot w , 0.35 , 0.75)$$

your weight $w \rightarrow$ your personal alert threshold t

context-dependent weights •
R1 • R5

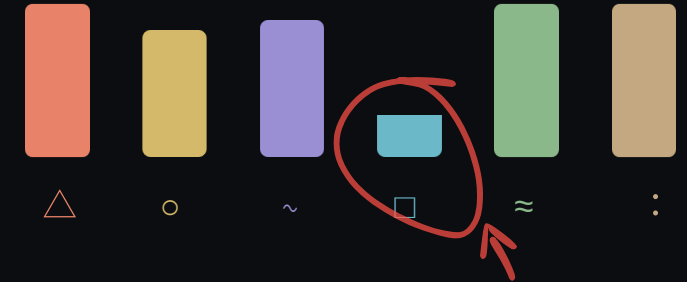
USER'S WEIGHTS



We don't average the senses

Averaging hides disasters. A room that's perfect in five senses and unbearable in one isn't "83% comfortable" — it's unbearable. So the overall score leans on the worst sense: a one-vote veto. You can't paper over a bad sense by being great elsewhere.

5 FINE • 1 FAILS



$$\text{overall} = (1-v) \cdot \text{mean_w} + v \cdot \text{worst} \quad (v = 0.5)$$

half the preference-weighted mean, half the single worst sense

non-additive IEQ comfort •
Yang & Moon 2019 (S2)

Senses don't live in separate boxes

“how a change to one sense **ripples** to another”

Soft surfaces absorb sound — a cosier floor makes a room quieter. A stuffy room feels hotter than it is. When one sense changes, Sensi nudges its partners and labels every nudge with why (research or physics). Discomfort spreads a little harder than comfort.



$\text{src} < 0.50 \rightarrow \text{partner} -0.06$ · $\text{src} \geq 0.70 \text{ (+edge)} \rightarrow +0.04$

a failing sense drags its partners; a strong one lifts them (capped)

multisensory thermal review
2025 · Spence 2020 (R2)

edges acoustic ↔ thermal · ventilation → olfactory · tactile → acoustic · glazing ↔ acoustic

+ personality introverts read “busy” as “too much”, so we tilt the stimulation senses: $\delta = 0.15 \cdot p \cdot (\text{stim} - 0.5)$

None of this is invented

The couplings come from peer-reviewed building science. We ran every claim through an adversarial fact-check — and threw out the ones that didn't survive (e.g. “acoustic always matters most”). Where evidence is thin, we encode the direction of an effect, not a fake-precise number.

18 / 25

claims confirmed by a 3-vote test

R1 autism sensory taxonomy · R2 multisensory mind · R3 cross-modal thermal · R5 arousal · R6 tactile warmth

S2 non-additive + acoustic ↔ thermal · S5 BS 8233 acoustic targets · S6 ASHRAE ventilation → olfactory

From a desktop plugin to the browser

WAS

MCP · ~~PyQt5~~ · ~~Grasshopper~~ / Rhino

NOW

FastAPI · React + Vite · LangGraph · in-process tools

26 nodes, verified

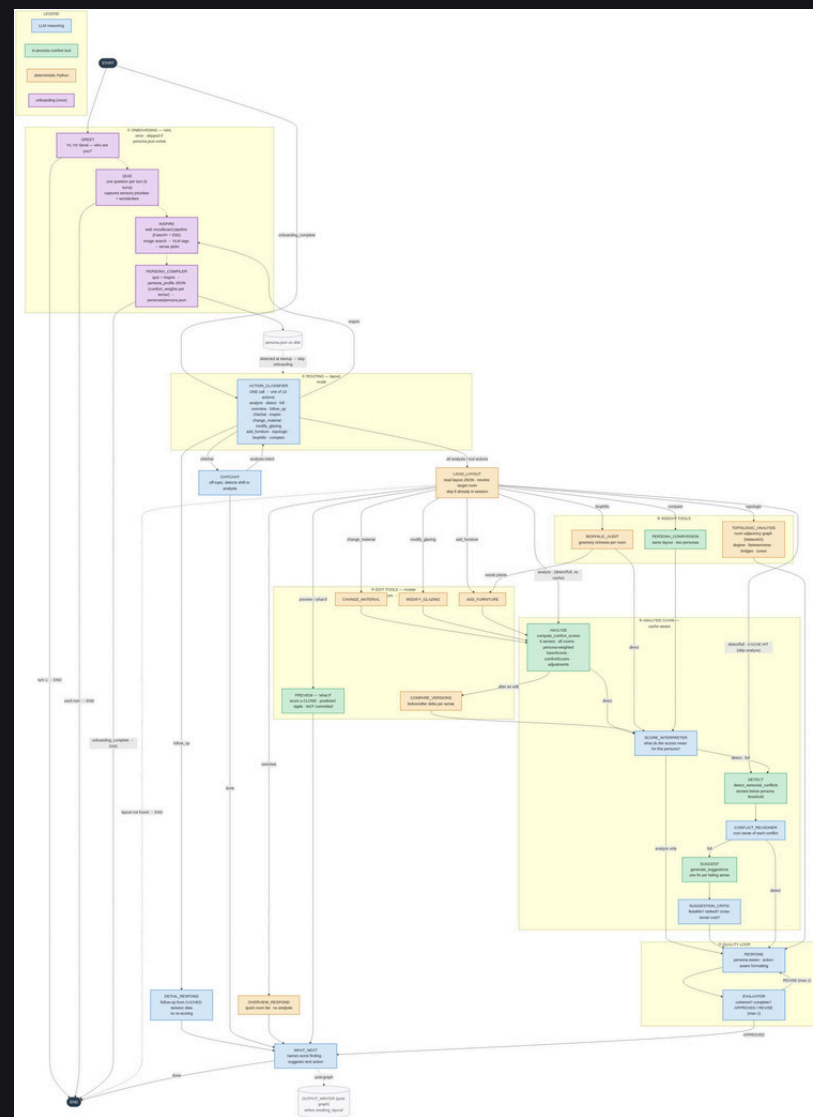
one action_classifier → 13 actions in one call

cache-aware re-scoring (skip redundant work)

edit → re-score → compare = the agentic ripple

WHAT THE COLOURS MEAN

- LLM reasoning
- in-process tool
- onboarding (once)
- Python step

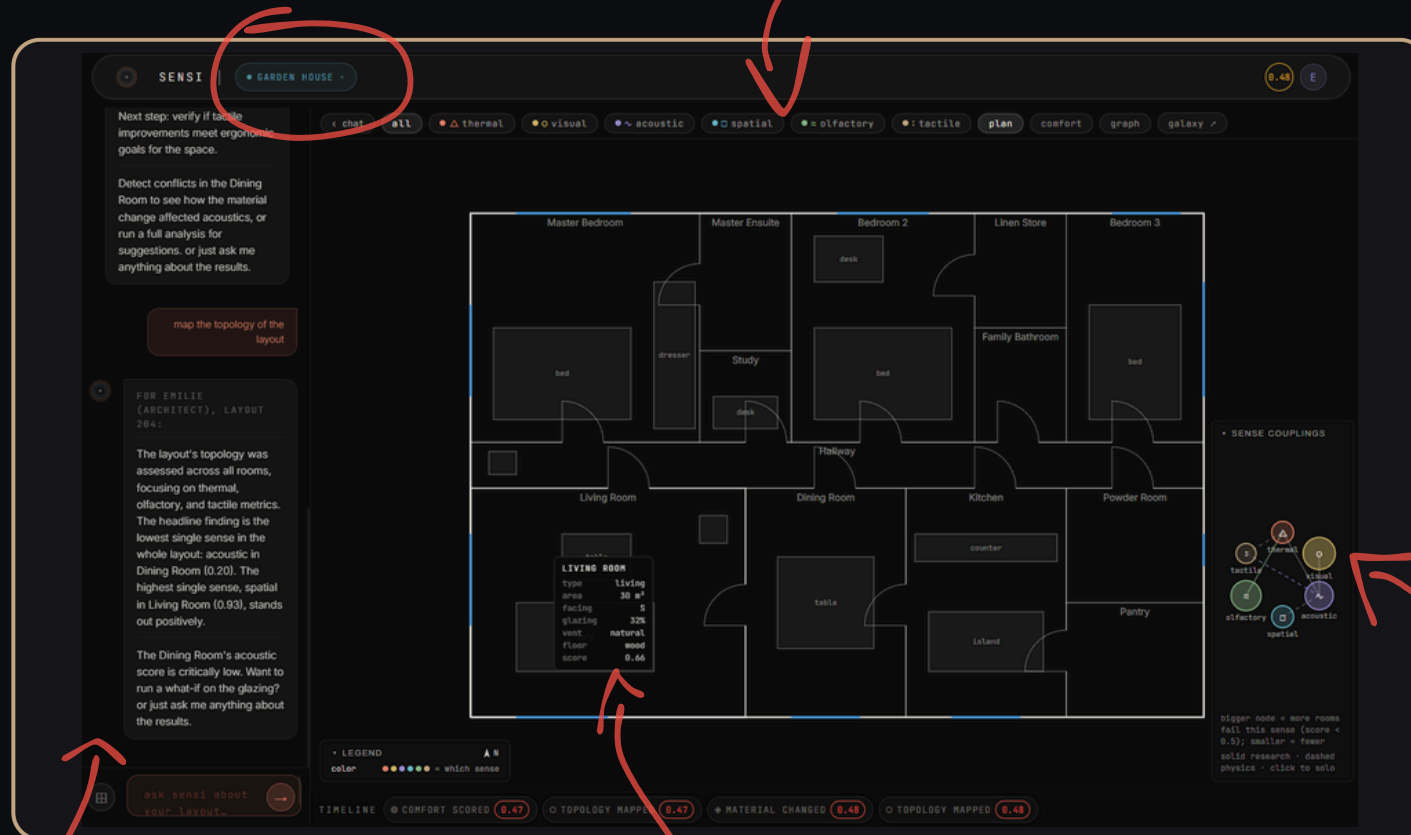


Rhino & Grasshopper are no longer required — Sensi runs anywhere, behind one link.

It starts with a layout

*pick a layout
or upload your own*

filtering and analysis options

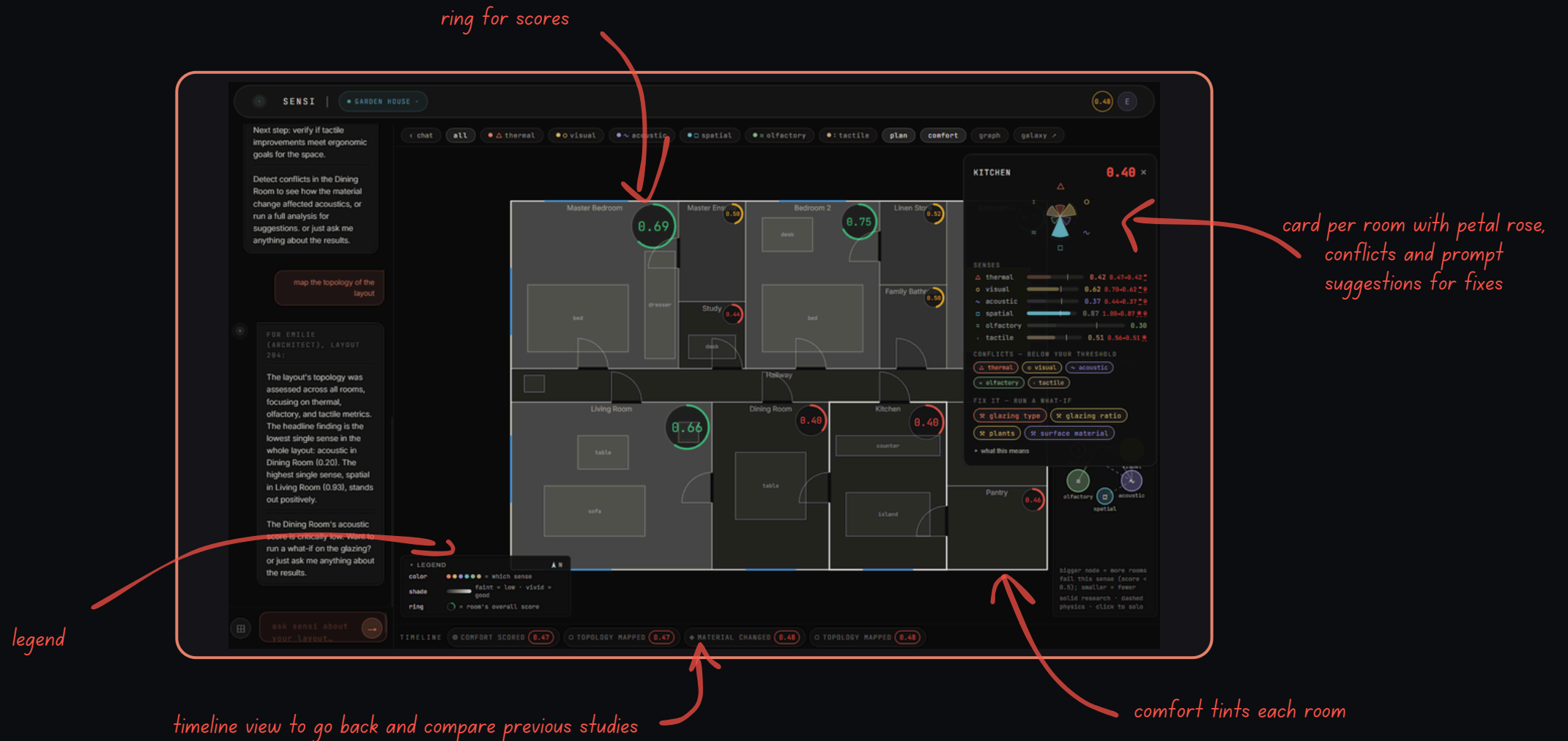


*key sense plan : relationship between
senses based on research and physics
and layout data*

expandable chat box

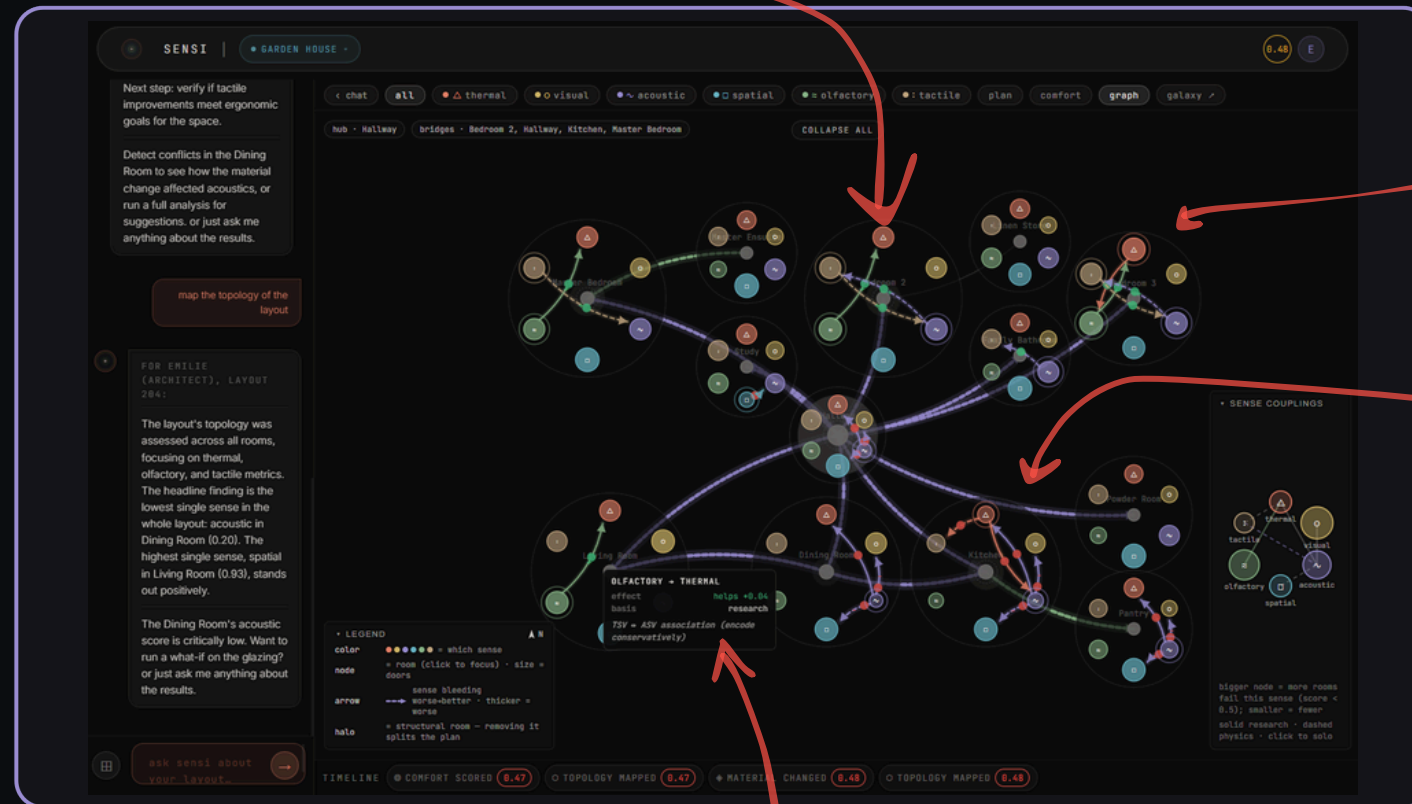
*loaded layout with data hover
feature*

Scores, made personal



Room topology = sensory zoning

sense couplings per room with ripple effect

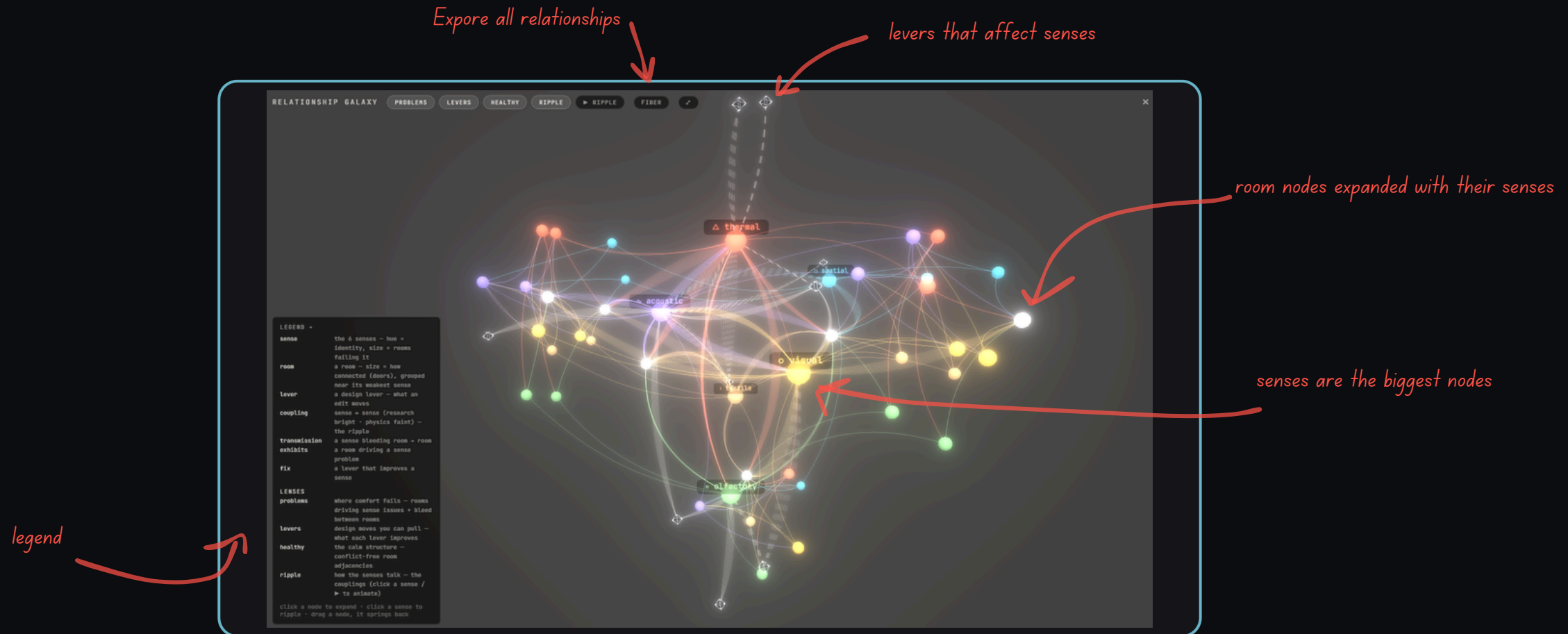


nodes = rooms
size of nodes = # of doors

arc = sense bleeding

hover on the elements to get more info on the relationship

The relationship galaxy



- ~ make the ripple more visual → pulses everywhere a value moves
- ◇ deepen the agentic loop → the agent proposes, previews, you accept
- Σ surface the formula → see & tune which coupling fired
- ✦ generative outputs → images of the improved space + more

every one makes the edges more legible.